

ADWAITH SANTHOSH

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Present Work Authorization (Authorized to work in the U.S F-1 OPT/CPT, STEM extension eligible)

EDUCATION

University of Maryland – College Park (College of Computer , Mathematics and Natural Sciences)

College Park, MD, USA

Master of Science (M.S), Data Science GPA: 4.0 (**Top 5 percentile**)

Aug 2024 – May 2026

• Relevant Coursework: Advanced Machine Learning, Deep Learning ,Natural Language Processing, Big Data Analytics,Probability and Statistics

WORK EXPERIENCE

AIB Inc

Dallas,TX, USA

Machine Learning Engineer Intern(CPT)

Jan 2025 – Present

• Building machine learning models for internal products for text based NLP classification, using Deep-learning and SVM models

CertbuddyAI Inc

Dallas,TX, USA

Machine Learning Engineer Intern(CPT)

June 2025 – Aug 2025

• Agentic Multimodal Conversation Bot Development: Developed an Agentic AI system with probabilistic and deterministic pipelines, enabling dynamic reasoning while ensuring consistent, fact-checked responses.

• Developed an OpenSearch vector DB based RAG engine with dynamic data partitions and S3 integration and efficiently indexing

University of Maryland - College Park(Department of Public Health)

College Park,MD, USA

Graduate Research Assistant

Oct 2024 –Jan 2026

• Bot Analysis Model Development: Leveraged a GNN-based model to accurately predict whether a Twitter post is made by a human or a bot, leveraging it for public health–related data analysis (research paper in progress).

• Developed a multimodal machine learning framework integrating text and image analysis for sentiment, hate, and anti-hate classification in social media memes, with applications to public health research.

GoML (Amazon Web Service GENAI Partner company)

Coimbatore, India

Machine Learning Engineer

Feb 2024 – Aug 2024

• RAG-based Chatbot Development: Developed a RAG-based chatbot for an EV Certification client, efficiently retrieving data from PDF files stored in Amazon S3 Bucket using QdrantDB and Claude 3.5 Sonnet with 95% accuracy ,reducing document retrieval time by 40%.

• System Architecture Design: Designed and developed the entire architecture utilizing AWS RDS Table, FastAPI, AWS Bedrock, AWS Lambda, AWS ECR, AWS EC2, and AWS S3 reducing latency by 35% and improving response time by 50%.

Tata Consultancy Service Limited

Chennai, India

Machine Learning Engineer

July 2021– Sep 2023

• Built and deployed a Deep Learning session-based recommender in Azure ML Studio via Pipelines, achieving <10 ms API latency with 10× response time reduction, and developed an AI-driven offer optimization algorithm boosting conversion rates by 20%.

• Optimized hybrid recommendation models (Matrix Factorization with LightFM) for a US-based retailer, cutting runtime from 7 hrs to

• 2.5 hrs (65% improvement) and automated Azure ML deployments to reduce manual effort by 40%.

TECHNICAL SKILLS

- **Programming Languages** – Python (Advanced), Java, Javascript ,C/C++/C#. **GEN AI & LLM Tools-** LangChain, Hugging Face,OpenAI,Claude.
- **DBMS** – SQL, Postgres, Oracle Database, MySQL, SQLite3, NoSQL, MongoDB, Firebase **Vector DB** - Pinecone,Weaviate,Qdrant,OpenSearch
- **Cloud Technologies** – Amazon Web Services (EC2, ECS, S3,RDS,Bedrock,Lambda), Kubernetes (GCP),Azure(ML Studio,AKS,ACI).
- **Machine Learning(ML)/Artificial Intelligence(AI)** – Deep learning (including LLM), Natural language processing (NLP), Big data, Neural networks, Machine learning algorithms, Linear algebra, Computer vision(image processing),CNN,RNN,LSTM, Reinforcement learning, Optimization clustering, Collaborative filtering, Classification techniques (Naïve Bayes, SVM, NN, KNN, Perceptron, PCA, Decision Tree), Feature selection, Vector distance/similarity, Data preprocessing, Data transformation, and Data visualization, Multimodal language models,Matrix factorization, Collaborative Filtering,Apache (Airflow,Hadoop,Spark),Terraform,Power BI,Tableau,GNN,Docker,Jupyter
- **Frameworks** – Flask, FastAPI,Django. **Performance Development Engineering** – Locust, Quantization and Distillation (to reduce latency).
- **Python Packages** – Scikit-learn, DuckDB, Spark (PySpark), Pandas, Numpy, Locust, TensorFlow, PyTorch,Spacy,NLTK,SciPy,Statsmodels .
- **Fundamentals** – Data structure and algorithms, Operating systems, Relational databases, Object-oriented programming/design, Agile methodology, Git, LINUX, Postman, Data science, Data processing, Parallel computing, Computer science, Computer engineering, Statistics, Data engineer, Computer architecture,Ci/CD Pipelines.**MLOps-** MLflow, Kubeflow,AzureML,SageMaker, Vertex AI, Databricks.
- **Others** – Articulate, Innovation, Innovative, Collaboration, Collaborative, Influence others, Written communication, Teamwork, Strategic, Problem solving, Organizational skills, Attention to detail, Analytical,Prompt engineering,Fine-tuning LLMs

ACHIEVEMENTS

- **Hackathon Winner-** CAFB Food security hackathon winner by building and deploying AI RAG chatbot in AWS that can create and ingest Docs.
- **Employee of the month** - Got employee of the month for May 2024 in GoML for outstanding product development.

RESEARCH & PUBLICATIONS

- A Decade of Discourse: Exploring Sentiments and Trends around Immigration on Social Media (2014–2024).(Big Data & Society)
- Developing an AI-powered Question-and-Answer Chatbot with English–Spanish Capabilities for New Mothers. (Big Data & Society)